PV102 Day 6 Student Packet

Inverter Charge Controller and Protection Boundaries

PV102 | CTS Day 22 | Student Print Packet
Power electronics set system boundaries. | Tuesday, August 11, 2026

Today's rule

Power electronics set system boundaries.

Learning targets

- Explain the roles of charge controllers and inverters in battery-based PV systems.
- Match inverter selection to continuous load, surge load, voltage, and output requirement.
- Identify where disconnects and overcurrent protection belong conceptually.
- Create a block diagram that separates source, storage, conversion, and load boundaries.

Morning quiz

1. An inverter primarily converts:

- A. DC to AC
- B. AC to sunlight
- C. wood to wire
- D. shade to volts

2. A charge controller protects and manages:

- A. Battery charging from PV
- B. Roof pitch
- C. Clipboards
- D. Nameplate stickers only

3. Inverter sizing must consider:

- A. Continuous load and surge
- B. Marker color only
- C. Roof direction only
- D. Lunch menu

4. An overcurrent device is used to:

- A. Open a circuit during excessive current
- B. Store solar energy
- C. Increase sunshine
- D. Convert AC to shade

5. A disconnect provides:

- A. A means to isolate equipment
- B. Free battery capacity
- C. More roof area
- D. A better compass

6. Why should source, storage, conversion, and load boundaries be shown separately?

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7. Name one label that belongs on a training system handoff diagram.

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8. Why can an inverter with enough continuous watts still fail a job?

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9. What is one thing students should not decide without manufacturer instructions?

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10. Write the Day 6 rule in your own words.

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Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Balance-of-System Block Diagram

Teams build a labeled block diagram and defend every protection/isolation point.

Scenario options

- 12 V training system with PV module, controller, LiFePO4 battery, inverter, DC load, AC load.
- 48 V bus system concept with hybrid inverter/charger, battery bank, PV strings, AC distribution.
- Remote DC-only sensor system with no inverter and limited service visits.

Team procedure

- Choose a scenario and place component cards in energy-path order.
- Draw source, storage, conversion, and load boundaries.
- Add conceptual OCPD/disconnect labels where energy could feed a fault or where service isolation is needed.
- Mark manufacturer-instruction-required decisions.
- Run a peer safety check: can someone tell what is energized, isolated, and unknown?
- Handoff the diagram and unresolved decisions.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

Which boundary on your diagram mattered most for safety, and what would happen if it were missing?

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